

Unsupervised Learning

Factor Analysis. PCA

Factor Analysis

Factor analysis is a method for investigating whether a number of features of interest $X_1, \dots, X_p$ (observable variables) are linearly dependent to a (smaller) number of unobservable factors $F_1, \dots, F_m$ (latent variables).

$$X_1 = \mu_1 + l_{11}F_1 + \dots + l_{1m}F_m + \epsilon_1$$

...

$$X_p = \mu_p + l_{p1}F_1 + \dots + l_{pm}F_m + \epsilon_p,$$

where

- $F_j, j = \overline{1, m}$ are the factors (components);
- l_{ij} is a **factor loading** indicating the influence that factor F_j has on the observable variable X_i .

Factor Analysis and PCA

Factor analysis:

- *Factor extraction*: transforms features into uncorrelated factors with corresponding loadings;
- *Factor rotation*: rotates the axes to make subsequent analysis easier.

Principal Component Analysis (PCA) is a factor analysis technique used to find *orthogonal* components and factor loadings so that each successive component contains the maximum possible variability.

The first principal component of the a set of features $X_1, \dots, X_p$ is the linear combination

$$Z_1 = \phi_{11}X_1 + \dots + \phi_{p1}X_p,$$

that has the largest variance and satisfies

$$\sum_{j=1}^p \phi_{j1}^2 = 1.$$

PCA

Let $\mathbf{X}$ be the $n \times p$ matrix encoding the data (each column corresponds to one of the p features). We assume that each variable in $\mathbf{X}$ is centered to have mean zero, i.e., each column of $\mathbf{X}$ has mean zero. We look for the linear combination of the sample features values,

$$z_{i1} = \phi_{11}x_{i1} + \dots + \phi_{p1}x_{ip}, \quad i = \overline{1, n},$$

that has largest variance subject to the constraint $\sum_{j=1}^p \phi_{j1}^2 = 1$. In other words, we determine the vector $\phi_1 = (\phi_{11}, \dots, \phi_{p1})^T$ that solves the following maximization problem

$$\max_{\phi_{11}, \dots, \phi_{p1}} \left\{ \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^p \phi_{j1} x_{ij} \right)^2 \right\} \quad \text{subject to} \quad \sum_{j=1}^p \phi_{j1}^2 = 1.$$

Geometric interpretation. The vector ϕ_1 defines a direction in feature space along which the data vary the most.

The proportion of variance explained (PVE)

- Total variance in the dataset (assuming the variables have been centered to have mean zero):

$$\text{Var}[\mathbf{X}] := \sum_{j=1}^p \text{Var}[X_j] = \frac{1}{n} \sum_{j=1}^p \sum_{i=1}^n x_{ij}^2.$$

- The variance explained by the m -th component Z_m :

$$\text{Var}[Z_m] := \frac{1}{n} \sum_{i=1}^n z_{im}^2 = \frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^p \phi_{jm} x_{ij} \right)^2.$$

- The PVE of the m -th principal component:

$$\frac{\text{Var}[Z_m]}{\text{Var}[\mathbf{X}]} = \frac{\sum_{i=1}^n \left(\sum_{j=1}^p \phi_{jm} x_{ij} \right)^2}{\sum_{j=1}^p \sum_{i=1}^n x_{ij}^2}.$$

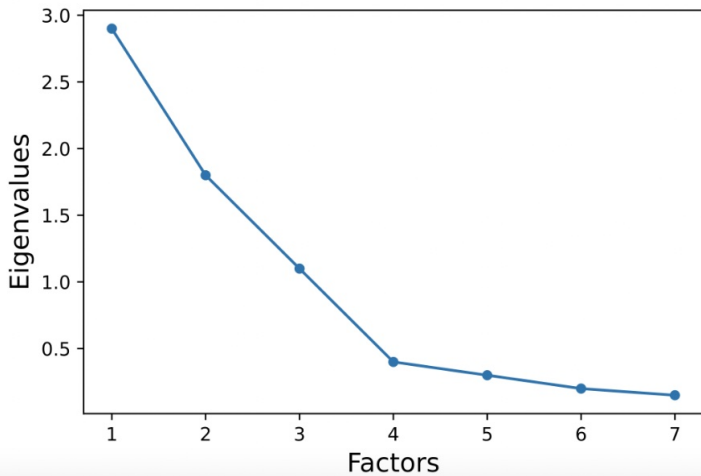
How many principal components to use?

- In theory, the same number of components can be used as the number of features. But some of the components can be discarded because of the limited variability those factors represent.
- Total variance in the dataset and the variance of the first M components.

$$\underbrace{\frac{1}{n} \sum_{j=1}^p \sum_{i=1}^n x_{ij}^2}_{\text{Var. dataset}} = \underbrace{\frac{1}{n} \sum_{m=1}^M \sum_{i=1}^n z_{im}^2}_{\text{Var. of first } M \text{ PCs}} + \underbrace{\frac{1}{n} \sum_{j=1}^p \sum_{i=1}^n \left(x_{ij} - \sum_{m=1}^M z_{im} \phi_{jm} \right)^2}_{\text{MSE of } M\text{-dimensional approx.}}$$

It follows that by maximizing the variance of the first M principal components, we minimize the MSE of the M -dimensional approximation.

Scree plots



More on PCA

- PCA is strongly connected to eigen decomposition.
(Think about $\mathbf{X}^T \mathbf{X} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^T$)
- **Centering and scaling the variables.** Before PCA is performed, the variables should be centered (to have mean 0) and scaled.